

TissuePMHC: A Matched-Pair Benchmark and Multi-Task Framework for Tissue-Specific MHC-I Presentation Preference

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Abstract—MHC-I presentation can vary among tissues even when the presenting allele and source protein are fixed. Existing predictors largely estimate tissue-blind peptide–MHC compatibility and therefore do not directly address this tissue-associated preference. We formulate a matched-pair ranking problem: given two peptides from the same parent protein under the same presenting restriction, determine which has stronger presentation evidence in a specified tissue. To remove a simple cross-tissue-frequency cue, the two peptides in every pair are additionally matched on their total number of positive tissue occurrences. The resulting benchmark contains 77 Human tissue–HLA tasks and 11 Mouse tissue–H-2 tasks, where H-2 is the mouse major histocompatibility complex. Both species use one fixed internal train/test split with 50 test pairs per task and the same three final-training seeds (20260704–20260706). Species-specific TissuePMHC configurations are selected by three-fold cross-validation confined to the training files before the fixed tests are opened. The occurrence audit finds zero mismatched pairs and identical positive-occurrence distributions between labels in both species and partitions. Tuned TissuePMHC reaches mean AUROC/AUPRC values of 0.8136/0.8126 on Human and 0.8194/0.8279 on Mouse. Tissue-independent external predictors remain near random in Human (AUROC 0.4874–0.5044) and Mouse (0.4917–0.5009). The tuning folds are not treated as strict-generalization estimates, and no peptide-separated evaluation is reported. These results assess tissue-conditioned peptide ranking for previously represented tissue–MHC tasks; they do not establish a causal tissue-processing mechanism, unseen-context generalization, independent-cohort validity, or clinical utility.

Index Terms—Tissue-specific antigen presentation, MHC-I, matched-pair benchmark, multi-task learning, peptide presentation.

1. Introduction

MHC-I molecules display intracellularly derived peptides at the cell surface for surveillance by cytotoxic CD8⁺ T cells. The resulting immunopeptidome is shaped by substantially more than peptide binding to an MHC: protein abundance and turnover, proteolytic degradation, antigen-processing pathways, cellular composition, tissue environment, and experimental detectability can all affect which peptides are observed. Multi-tissue immunopeptidomic studies have consequently revealed both widely shared ligands and pronounced tissue-associated differences [2, 3]. Nevertheless, most computational models are designed to predict peptide binding to or presentation by an MHC in a general setting and therefore do not directly characterize how presentation evidence varies across tissues under a fixed MHC context.

Tissue-associated presentation preference by MHC-I can arise at several successive stages of antigen processing. Genes encoding constitutive or immunoproteasomal sub-

units, TAP1/2 transporters, ERAP1/2 aminopeptidases, and components of the peptide-loading machinery are not expressed uniformly across tissues. Together with tissue-dependent source-protein abundance and cell-type composition, these differences can alter which proteins are cleaved, which peptide fragments enter the endoplasmic reticulum, how precursor peptides are trimmed, and which sufficiently stable peptide–MHC-I complexes are ultimately displayed at the cell surface [4]. Thus, the same source protein may generate different presented epitopes in different tissues, while the same epitope may differ across tissues in the amount or likelihood of surface presentation. Understanding this variation is important for cancer therapy: it can help prioritize epitopes that are efficiently presented by tumors but minimally presented by essential normal tissues, thereby improving target selection and reducing potential on-target, off-tumor toxicity. In the present study, these mechanisms provide a biological explanation for tissue-associated preference rather than a causal claim, because the model does not directly measure gene expression or the activity of individual processing steps.

Figure 1 summarizes these mechanisms and the biological motivation for tissue-conditioned presentation.

Existing tools answer related but different questions. Some estimate whether a peptide can bind an MHC; others estimate whether it is likely to be presented in a pooled or sample-specific setting, sometimes using expression or processing information [6, 7, 9]. Here we ask a narrower relative question: when the MHC allele and source protein are held constant, which of two peptides has stronger evidence in one specified tissue? Holding these two factors constant reduces easy explanations based only on peptide–MHC compatibility or protein identity and makes tissue-associated differences the focus of the comparison.

We call this question tissue–MHC-specific presentation preference prediction. Each prediction context corresponds to one tissue and one MHC restriction. Let p denote a peptide and let $\tau = (t, m)$ denote that tissue–MHC combination. A model produces a score

$$s_{\theta}(p, \tau) \in \mathbb{R},$$

which is used to rank peptides within that combination. A larger score means stronger predicted evidence relative to other candidates in the same context; it is not an absolute probability that the peptide is present. The model is evaluated only for tissue–MHC combinations represented during training, so it should not be interpreted as a predictor for a new tissue, a new MHC allele, or a new tissue–MHC combination.

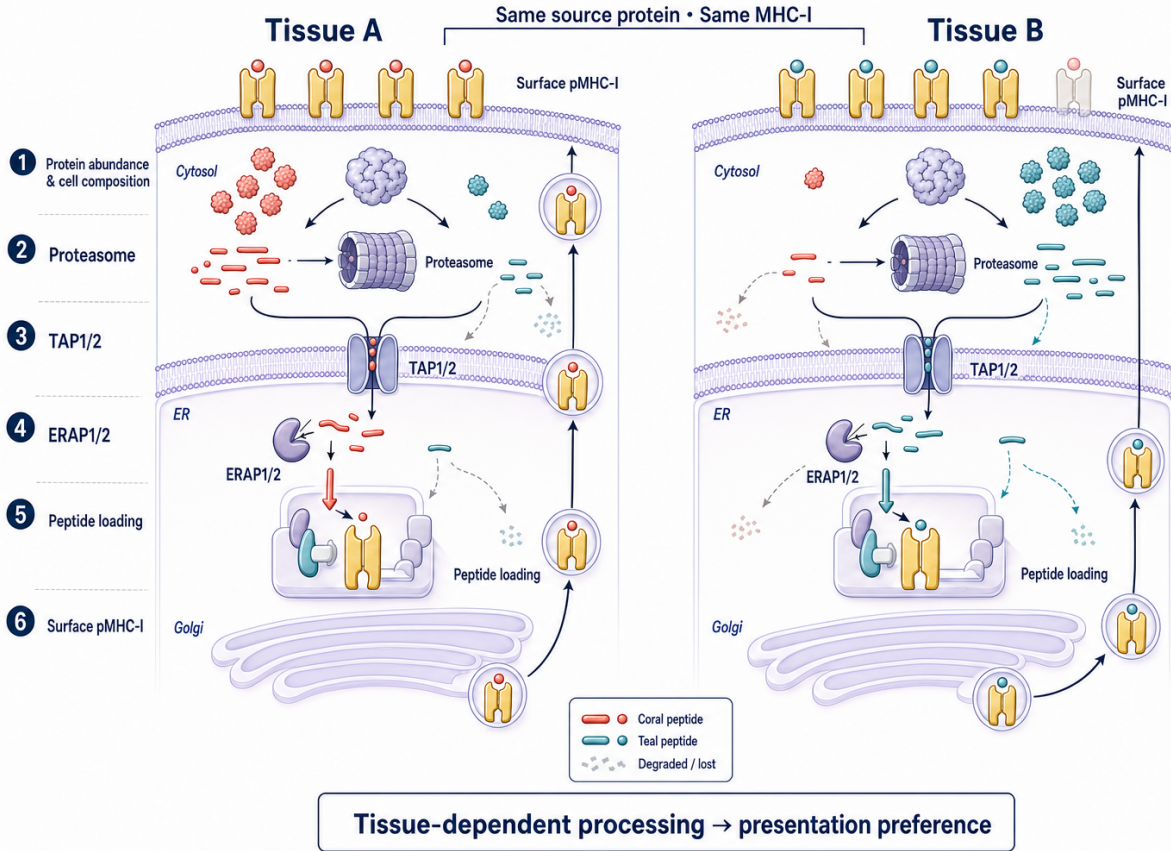


Figure 1: Conceptual mechanisms that can generate tissue-associated presentation preference by MHC-I under a fixed source-protein and MHC context. Tissue-dependent source-protein abundance and cell-type composition, TAP-dependent transport, ERAP-mediated trimming, and peptide loading can enrich different peptides for surface display. Dashed paths denote peptide degradation or loss from the presented repertoire, not confirmed biological absence. The schematic motivates an association and does not establish causality for any individual processing step.

To construct training examples aligned with this objective, each recorded-positive peptide is paired with a comparison peptide that has evidence elsewhere but not in the target tissue (a pseudo-negative peptide) under the constraints

$$m^+ = m^-, \quad u^+ = u^-,$$

where m is the MHC restriction and u is the parent UniProt protein. One peptide is reported in the target tissue; its comparison peptide is reported elsewhere under the same MHC and from the same protein, but not in the target context. Biologically, the pair asks whether two fragments of one protein show different tissue-associated evidence after controlling the presenting allele and source-protein identity. The second peptide is a pseudo-negative rather than a confirmed biological absence. Missing reports can reflect incomplete sampling or limited mass-spectrometry detection, and the pairing cannot remove study, batch, expression, or cell-composition effects.

On this basis, we introduce paired human and mouse TissuePMHC benchmarks and separately trained species-specific prediction models. Each model combines a globally shared peptide representation with a representation restricted to one MHC allele; in biological terms, this allows broadly shared sequence patterns and allele-associated motifs to con-

tribute separately. To prevent total cross-tissue reporting frequency from separating the labels, the two members of every pair have the same number of positive tissue occurrences. Evaluation uses one fixed internal train/test split per species, with 50 test pairs per retained tissue–MHC combination and three independent final-training seeds. Species-specific hyperparameters are selected by cross-validation confined to the training split before the fixed test is opened. A held-out peptide is absent from fitting rows for the same combination but may recur in another combination.

The study makes four main contributions. First, it defines a biologically motivated comparison in which MHC restriction, parent protein, and total positive tissue-occurrence count are matched. Second, it provides Human and Mouse benchmarks with the same fixed-test allocation of 50 pairs per retained tissue–MHC combination. Third, it verifies pairwise and taskwise that occurrence-count distributions are identical between labels, making this count alone exactly non-discriminative. Fourth, it tests whether the retained signal depends on tissue identity or can be explained by general peptide–MHC compatibility alone. The aim is therefore not only to report a score, but also to state which shortcut has been controlled and which observational limitations remain. After the background, the paper proceeds from data and task

definition to balance and leakage audits, the model and evaluation protocol, primary Human and Mouse results, strict-generalization scope, component and mechanism-oriented comparisons, tissue and allele heterogeneity, and finally external controls.

2. Related Work

2.1. Peptide–MHC Binding and Presentation Prediction

Peptide–MHC binding prediction is a foundational problem in computational immunology. Established methods such as NetMHCpan [6], MHCflurry [7], and ACME [8] estimate binding affinity, binding rank, or peptide–MHC compatibility from peptide sequences and allele information. Pan-allelic predictors improve coverage by representing MHC molecules through pseudo-sequences that summarize the peptide-binding groove or through related allele features. This representation allows information sharing across well-characterized and data-scarce alleles [5, 6]. These models are essential for filtering candidate epitopes, but binding is only one stage of antigen presentation. A peptide with favorable binding characteristics may still be absent from the cell-surface immunopeptidome because its source protein is not expressed or degraded, the peptide is not generated or transported efficiently, or it is not detected experimentally. Binding scores therefore do not directly encode presentation-evidence preferences among tissues that share the same MHC restriction.

2.2. Predicting Cell-surface Antigen Presentation (Antigen Presentation Prediction)

Eluted-ligand data enable models to learn a broader presentation signal than binding measurements alone [6, 7]. General presentation predictors combine peptide sequence, MHC context, and, in some cases, antigen-processing or source-protein features to distinguish observed ligands from background peptides. Systems such as BigMHC [9] illustrate how large immunopeptidomic collections and modern sequence models can improve ligand presentation prediction and candidate ranking. Their usual target, however, is whether a peptide is likely to be presented for an allele, sample, or pooled presentation setting. This target differs from the relative question studied here: given a particular tissue–MHC context, which of two peptides from the same parent protein has stronger tissue-conditioned presentation evidence? Consequently, a high general-presentation score need not imply a tissue-specific preference, and direct comparisons require a benchmark whose supervision reflects that distinction.

2.3. Tissue- and Sample-Context Immunopeptidomics

Multi-tissue immunopeptidomic atlases have revealed that ligand repertoires contain both broadly shared peptides and tissue-associated components [2, 3]. MHCflurry 2.0 incorporates antigen-processing features but does not use tissue identity [7]. More recent systems perform multi-allelic deconvolution, as in ImmuneApp [20], or combine peptide, MHC, peptide-flanking sequence, and optional gene-expression features, as in HLApollo [21]. These approaches demonstrate that presentation is context dependent, but they generally do not define a tissue–MHC combination as the su-

pervised prediction unit or match every recorded positive and pseudo-negative on both MHC restriction and parent protein. Our formulation is therefore complementary to expression- and sample-aware prediction: it isolates a ranking problem conditioned on a known biological context that can later be combined with molecular measurements.

2.4. Sharing Information across Tissue–MHC Combinations

Some tissue–MHC combinations contain many observations, whereas others contain relatively few. Learning a completely separate model for every combination would waste information shared across tissues and alleles. Conversely, pooling all combinations into one model could blur allele-specific binding motifs and tissue-associated differences. Joint learning provides a middle ground: related combinations share a peptide representation while retaining separate prediction functions [10, 11]. More structured methods allow different groups to use shared and specialized model components [12, 13]. TissuePMHC adopts a biologically structured sharing scheme: one model view learns patterns across all human combinations, while a second shares information only among tissues with the same HLA allele.

2.5. Relation to This Work

The principal distinction from previous work is the prediction target. General predictors estimate peptide binding to or presentation by an MHC, whereas TissuePMHC ranks peptides matched by MHC restriction, parent protein, and total positive tissue-occurrence count within a specified tissue–MHC context. The human and mouse datasets define this comparison identically, although each species uses a separately trained model. The current occurrence-matched experiments use a fixed internal test set and allow a peptide to recur in another fitting combination. MHCflurry and NetMHCpan are therefore used as tissue-independent peptide–MHC controls rather than as tissue-aware competitors.

These distinctions are stated in prose because the published methods use different targets, negative constructions, datasets, and evaluation protocols. NetMHCpan and MHCflurry provide tissue-independent peptide–MHC controls in the present fixed test; BigMHC, ImmuneApp, and HLApollo remain related presentation systems rather than directly comparable tissue-aware baselines. The controlled architecture surveys below therefore contain only systems evaluated on the same occurrence-matched rows, task inventory, and fixed split.

3. Data and Task Definition

3.1. Defining a Tissue–MHC Prediction Context

One prediction context corresponds to one tissue and one MHC-I restriction. For example, lung with HLA-A*02:01 and liver with the same allele are different contexts because the tissue differs. Let t denote a tissue and m an MHC-I restriction. We define the set of observed contexts as

$$\mathcal{T} = \{\tau = (t, m)\}.$$

Each benchmark row is represented by

$$x_i = (p_i, \tau_i), \quad y_i \in \{0, 1\},$$

where p_i is a nine-residue peptide and τ_i identifies one tissue–MHC context. A positive label, $y_i = 1$, indicates positive ligand evidence in the target context. A comparison label, $y_i = 0$, indicates positive evidence under the same MHC restriction and parent protein in at least one other tissue, but no report in the target context. Thus, the pseudo-negative label represents absence of recorded evidence rather than experimentally verified non-presentation.

Given (p, τ) , a model parameterized by θ produces

$$s_\theta(p, \tau) \in \mathbb{R},$$

with larger values indicating stronger predicted evidence within tissue–MHC combination τ . Operationally, the model learns to rank a peptide reported in the target tissue above a comparison peptide matched by MHC restriction and parent protein. It is trained on individual peptide rows, while the paired construction controls example generation and evaluation. Because each tissue–MHC combination has its own prediction function, the benchmark evaluates previously represented combinations; it does not assess unseen tissues or unseen MHC restrictions.

3.2. Pairing Peptides by Tissue, MHC Restriction, and Parent Protein

Each pair j contains one positive peptide p_j^+ and one pseudo-negative peptide p_j^- with evidence elsewhere but not in the target tissue. Let $c(p, m, u)$ be the number of tissues with positive source-record evidence for peptide p under MHC restriction m and parent protein u . The two rows share the same target tissue, MHC restriction, parent UniProt protein, and positive tissue-occurrence count:

$$t_j^+ = t_j^-, \quad m_j^+ = m_j^-, \quad u_j^+ = u_j^-.$$

and

$$c(p_j^+, m_j, u_j) = c(p_j^-, m_j, u_j).$$

This matching removes three easy explanations for a difference: the peptides cannot be separated merely because they use different MHC molecules, come from different proteins, or have different total numbers of positive tissue reports. The remaining difference can include tissue-associated processing and presentation, but also the identities of the reporting tissues, cell composition, study design, detection sensitivity, and annotation completeness. The benchmark therefore measures tissue-associated evidence, not a purified molecular mechanism.

We first calculate performance separately for every tissue–MHC combination and then give each combination equal weight, so common combinations do not dominate rare ones. We also report within-pair ordering accuracy:

$$\text{Acc}_{\text{pair}} = \frac{1}{N} \sum_{j=1}^N \mathbf{1} \left[s_\theta(p_j^+, \tau_j) > s_\theta(p_j^-, \tau_j) \right].$$

where N is the number of held-out pairs. This quantity is the fraction of pairs for which the peptide reported in the target tissue receives the higher score. AUROC measures ordering across all rows within a tissue–MHC combination, whereas within-pair accuracy directly evaluates the original MHC- and parent-matched comparison.

3.3. Evidence Inclusion Criteria

The human and mouse benchmarks are derived from the full ligand export of the IEDB downloaded on July 4, 2026 [1]. We apply the same biological inclusion rules to both species. We retain only clearly positive ligand records for MHC-I with an identifiable parent protein and a standard unmodified 9-amino-acid peptide:

- a qualitative measurement whose normalized text begins with *positive*;
- a MHC-I restriction, with both peptide source and host matching Homo sapiens for the human benchmark or Mus musculus for the mouse benchmark;
- a human restriction matching the four-digit HLA pattern HLA-...*dd:dd (the extractor also accepts the equivalent colon-free four-digit form) or a restriction from the mouse major histocompatibility complex (H-2; written as H2 in the processed identifiers) beginning with H2-;
- a parent UniProt accession extracted from the molecule-parent web identifier (internationalized resource identifier; IRI) in the IEDB;
- a linear, unmodified peptide containing only the 20 standard amino acids;
- a peptide length of exactly nine residues; and
- deduplication before pairing by the exact peptide, restriction, tissue, and parent/source-protein annotation fields.

Source-tissue values are stripped of surrounding whitespace and retained as the exact names used in the IEDB; distinct nonempty tissue strings are not merged after extraction. Empty or missing-value-like fields required to define a tissue–MHC combination are removed before benchmark filtering. Pair-consistency checks require both rows to have the same nonempty tissue and restriction. Retained HLA and H-2 strings are used directly as identifiers after the pattern checks above. Records without a parent UniProt accession are excluded because parent-protein matching is a defining control of the benchmark. Restricting the study to unmodified canonical 9-residue peptides for MHC-I provides a consistent sequence space, but excludes variable-length ligands, post-translationally modified peptides, and presentation by MHC-II.

3.4. Constructing MHC- and Parent-matched Peptide Pairs

For each target tissue–MHC combination, we compare a peptide reported in that tissue with another peptide from the same protein that is reported in a different tissue under the same MHC restriction. A fixed random state, 20260704, makes the following construction reproducible:

1. Group all eligible ligand records by MHC restriction m and parent UniProt protein u .
2. For each target combination (t, m, u) , collect peptides with positive evidence in tissue t and calculate their positive tissue-occurrence counts.

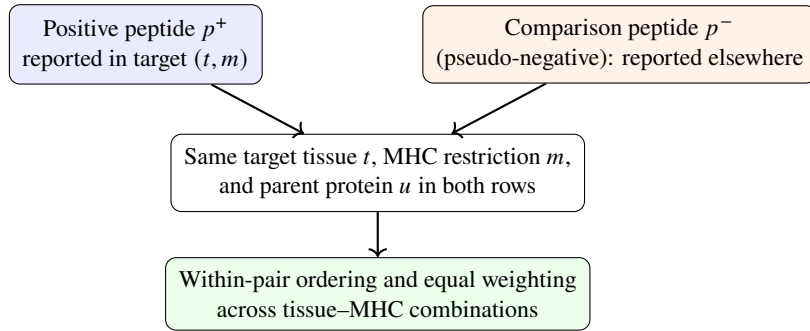


Figure 2: Construction of a peptide pair matched by target tissue, MHC restriction, and parent protein. The comparison peptide has positive evidence elsewhere but no report in the target tissue-MHC context.

3. From other tissues represented in the same (m, u) group, collect candidate comparison peptides with evidence elsewhere but not yet in the target tissue (pseudo-negative peptides), stratified by the same occurrence count.
4. Exclude every candidate with a positive report in the target tissue-MHC combination (t, m) .
5. Within each target combination, parent protein, occurrence-count stratum, and label, use each peptide at most once. Set the number of pairs to the smaller of the eligible positive and pseudo-negative counts and sample both sets without replacement.
6. Randomly permute the selected pseudo-negatives and join them one-to-one with the selected positives.
7. Assign a persistent pair identifier, preserve the reported-tissue and protein annotations, and verify that every identifier has exactly one row per label, identical tissue, restriction, and parent UniProt values in both rows, and no target-tissue report for the pseudo-negative.

Every pair contributes one target-tissue peptide and one comparison peptide, so the dataset is deliberately balanced. This balance is useful for controlled ranking but does not resemble the natural frequency of presented peptides in a tissue sample. The reported scores therefore describe relative ranking within this constructed benchmark, not the probability that a peptide would be detected in a biological specimen.

Figure 2 summarizes the matched-pair unit used throughout the occurrence-matched benchmark.

3.5. Human Benchmark

The occurrence-matched human benchmark contains 77 tissue-HLA combinations. For every matched pair, the positive and pseudo-negative peptides have the same number of positive tissue occurrences in the source records. This additional control prevents the model from separating the labels merely by learning that one member of a pair was reported in more tissues overall. Each retained combination contributes exactly 50 complete pairs to the fixed internal test set, and all other pairs form the training partition. Random state 20260704 makes this division reproducible.

Table 1 summarizes the Human task inventory and fixed split.

Table 1: Occurrence-matched human benchmark and fixed internal test-set statistics.

Property	Value
Tissue-HLA combinations	77
Tissues	13
HLA alleles	29
Minimum retained pairs per combination	100
Training pairs	21,489
Test pairs	3,850
Training rows	42,978
Test rows	7,700
Test pairs per combination	50

The 77 tissue-HLA combinations form an incomplete and long-tailed tissue-by-HLA coverage matrix. Performance is therefore calculated separately for each combination before averaging with equal weight; pooling all rows would overemphasize combinations with more examples and could conceal weak performance in sparsely represented contexts. The fixed internal test set excludes complete pair identifiers, but a peptide or parent protein may still occur in a different combination on both sides of the split. It therefore evaluates familiar tissue-HLA contexts rather than biological entities that are entirely absent from training.

3.6. Mouse Benchmark

The mouse benchmark uses the same tissue-conditioned and occurrence-matching definitions: the two peptides share their H2 restriction, parent protein, and total number of positive tissue occurrences. The supplied occurrence-matched files retain 11 tissue-H2 combinations. Exactly 50 complete matched pairs per retained combination are assigned to the fixed internal test set using random state 20260704, and the remainder form the training partition.

Table 2 reports the corresponding Mouse task inventory and fixed split.

The benchmarks are intentionally parallel in construction but not equal in scale. After applying the same biological filters and occurrence-matching rule, the human files contain 77 retained tissue-HLA combinations, 13 tissue labels, 29 HLA alleles, and 25,339 pairs, whereas the mouse files contain 11 tissue-H2 combinations, four tissues, four H2 restrictions, and 2,639 pairs. The fixed test allocation is identical—50 pairs per retained combination—so the test-set difference

of 3,850 versus 550 pairs follows exactly from the 77-versus-11 task inventories. The remaining difference reflects the number and coverage of eligible source records and retained tissue–MHC contexts, not different per-task sampling rules. Consequently, raw performance differences between species are descriptive and cannot be interpreted as species effects.

4. Data Balance and Leakage Audit

4.1. Occurrence-balancing Audit

We audited the supplied training and test files pair by pair before model evaluation. A valid pair must contain exactly one row of each label; both rows must have the same tissue, MHC restriction, and parent UniProt identifier; and the two peptides must have the same `presentation_tissue_count`. All 27,978 pairs across the two species satisfy these conditions, with zero mismatched pairs in either partition. The complete Human benchmark contains 32,625 positive tissue occurrences in each label class, and the complete Mouse benchmark contains 2,807 in each label class. Equality also holds separately within every task in the training and test partitions (Table 3), including the complete frequency distribution rather than only its total. A score using only `presentation_tissue_count` has AUROC 0.5000 in every task (minimum, mean, and maximum all 0.5000). Thus, label identity cannot be inferred from total positive tissue-occurrence count in this benchmark.

Controlled shortcut versus residual sources of bias. We use occurrence matching to remove one prespecified shortcut: a model cannot favor the label-1 peptide simply because it was reported positively in more tissues overall. This result should not be described as removal of all dataset bias. The construction does not balance the identities of the other reporting tissues, study or laboratory provenance, donor composition, tissue coverage, allele coverage, sample preparation, mass-spectrometry platform, detection depth, or annotation completeness. It also does not prevent an exact peptide or parent protein from recurring in another tissue–MHC combination. Statements about bias in the remainder of the paper therefore distinguish the eliminated occurrence-count shortcut from these residual observational and provenance effects.

4.2. Entity Overlap across the Fixed Split

In both species, a held-out peptide is absent from the training rows of the tissue–MHC combination being evaluated, but it may occur in training rows for another combination. Table 4 makes this distinction explicit. For each test row, the audit first asks whether the same entity occurs in training for the current combination and, only if it does not, whether it occurs in another combination. The three columns are therefore mutually exclusive.

Table 4 answers an entity-reuse question about the saved train/test split; Table 5 instead defines the stages used for training-only model selection, final fitting, and fixed-test es-

Table 2: Occurrence-matched mouse benchmark and fixed internal test-set statistics.

Property	Value
Tissue–H2 combinations	11
Tissues	4
H-2 restrictions	4
Minimum retained pairs per combination	134
Training pairs	2,089
Test pairs	550
Training rows	4,178
Test rows	1,100
Test pairs per combination	50

Table 3: Occurrence-balancing audit. Occurrence totals sum `presentation_tissue_count` separately over the two label classes. A mismatch denotes a pair whose two rows have different occurrence counts.

Species	Partition	Pairs	Label 0/1 totals	Mismatches
Human	Training	21,489	27,706/27,706	0
Human	Test	3,850	4,919/4,919	0
Mouse	Training	2,089	2,204/2,204	0
Mouse	Test	550	603/603	0

timation. No separate cross-validation overlap table is reported because the tuning folds are not a generalization result and would duplicate the pair-grouping rule already stated in the evaluation protocol.

Thus, each test peptide is new to the specific tissue–MHC task being evaluated, although 65.35% of Human and 64.36% of Mouse test-row peptides occur in training for another task. Parent proteins recur more often, including within the current task. The fixed test is therefore an internal evaluation in familiar contexts, not an entity-disjoint or independent external validation.

5. Model and Evaluation Protocol

5.1. Model Overview and Tissue–HLA-specific Prediction

Peptide presentation contains sequence patterns at more than one biological level. Some features may recur across many tissues and HLA alleles, whereas others may reflect motifs associated with one allele. TissuePMHC therefore contains a global pathway that learns from all tissue–HLA combinations with additional tissue and allele supervision, and an HLA-specific pathway that learns only from combinations sharing one HLA allele. Each pathway produces a separate score for every tissue–HLA combination. Human and Mouse use separately tuned configurations selected by cross-validation confined to the corresponding training split. Both final models are fitted with the same three random initializations; unless an ensemble is explicitly identified, results summarize seed-level metrics rather than averaging row-level predictions.

Both pathways use a position-aware peptide encoder that preserves residue position and examines neighboring residues, followed by a separate output layer for each observed tissue–HLA combination. This structure allows combinations with many examples to support those with fewer examples without forcing all alleles to use an identical peptide representation. Because the final prediction depends on an output layer learned for a particular combination, the model is not designed for tissue–HLA combinations absent from training.

Figure 3 summarizes the two sharing pathways and their fixed-test score combination.

The remainder of this section gives the exact implementation needed for reproduction. Let h_i denote the numerical representation of peptide p_i , and let τ_i denote its tissue–HLA combination. Each pathway maintains a separate linear output for every available combination. The corresponding logit is

$$z_i = g_{\tau_i}(h_i),$$

and the associated presentation-preference probability is

$$q_i = \sigma(z_i),$$

where σ is the logistic sigmoid. During each mini-batch update, examples are routed to the output layer for their tissue–HLA combination, while encoder parameters are shared according to the pathway definition. The main objective is binary cross-entropy with logits, denoted BCE:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{B} \sum_{i=1}^B [y_i \log q_i + (1 - y_i) \log(1 - q_i)].$$

Although every benchmark pair contains one recorded-positive row and one pseudo-negative row, the model processes rows independently. Pair identifiers are not used during the forward pass or loss calculation.

5.2. Peptide Encoding That Preserves Position and Examines Several Motif Lengths

The biological meaning of an amino acid depends on its position in a 9-mer peptide: anchor residues involved in MHC binding are not interchangeable with residues at other positions. The encoder therefore preserves all nine positions while examining local patterns of several lengths. For a peptide $p = (a_1, \dots, a_9)$, each residue is mapped to a learned 16-dimensional embedding and combined with a learned positional embedding:

$$X = E(p) + P \in \mathbb{R}^{9 \times d},$$

where $E(p)$ contains the residue embeddings and P is a trainable $9 \times d$ positional representation shared across peptides. These positional embeddings distinguish the same amino acid at different locations, which is important because anchor and supporting positions in MHC-I ligands are not interchangeable.

Table 4: Human and Mouse entity-overlap audit for the occurrence-matched fixed internal test sets. Counts are test-row occurrences, except for pair identifiers, which are counted once per pair. “Current combination” means the tissue–MHC combination of the test row.

Species and held-out entity	Current-combination training	Other-combination training only	Absent from all training
Human pair identifier (3,850 pairs)	0 (0.00%)	0 (0.00%)	3,850 (100.00%)
Human peptide identity (7,700 rows)	0 (0.00%)	5,032 (65.35%)	2,668 (34.65%)
Human parent UniProt (7,700 rows)	1,112 (14.44%)	6,152 (79.90%)	436 (5.66%)
Mouse pair identifier (550 pairs)	0 (0.00%)	0 (0.00%)	550 (100.00%)
Mouse peptide identity (1,100 rows)	0 (0.00%)	708 (64.36%)	392 (35.64%)
Mouse parent UniProt (1,100 rows)	180 (16.36%)	726 (66.00%)	194 (17.64%)

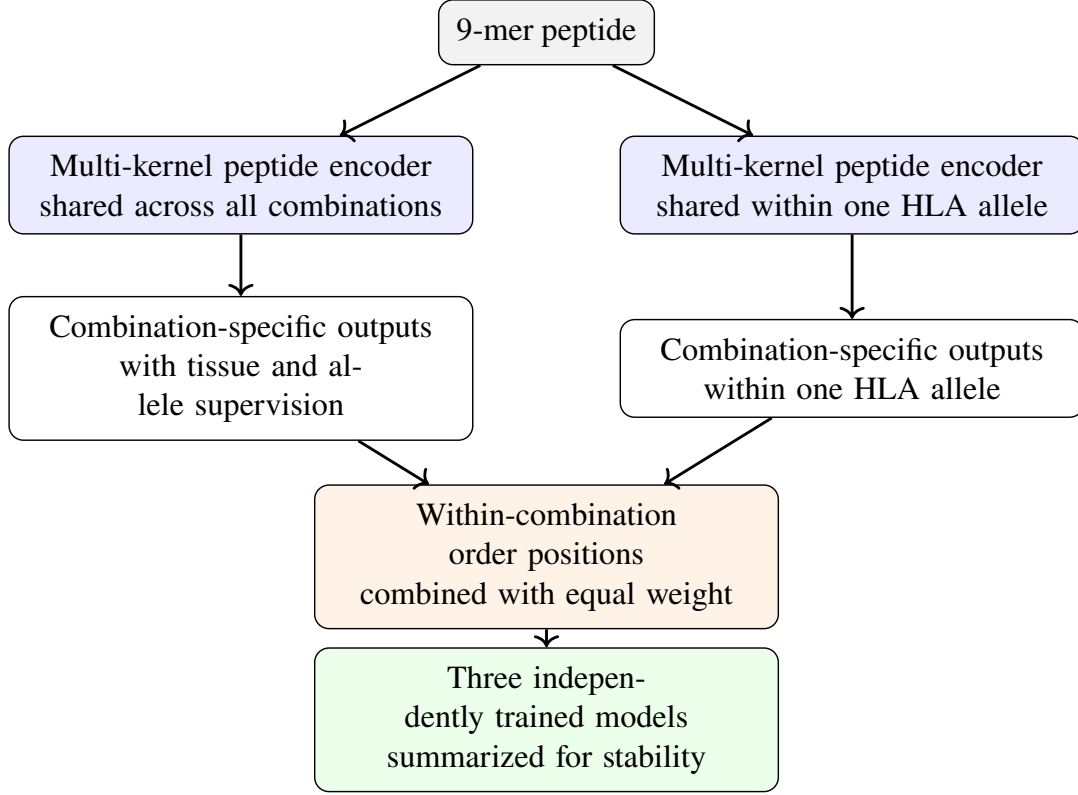


Figure 3: Overview of the human model. The global pathway learns peptide patterns across all tissue–HLA combinations, whereas the HLA-specific pathway shares information only among tissues with the same allele. Tissue and allele prediction provide training supervision but do not enter the final score. Each training run combines the two within-combination rankings; three seeds quantify optimization stability.

The model operates only on canonical 9-mer peptides. No tissue transcriptomics, proteomics, parent-protein sequence, flanking sequence, or external binding score is supplied as input. Tissue and HLA information enter through the combination identifiers, combination-specific output layers, parameter-sharing structure, and auxiliary training labels rather than through additional omics measurements.

To examine local sequence patterns spanning two, three, or five residues, we apply one-dimensional convolutions with widths

$$\mathcal{K} = \{2, 3, 5\}.$$

For each $k \in \mathcal{K}$, the input is transposed to a channels-first representation and processed as

$$H_k = \text{ReLU}(\text{Conv1D}_k(X)).$$

Each convolution produces 32 feature maps. Padding retains local context at the peptide ends. For even kernel widths, symmetric padding produces one additional position; the final position is therefore cropped to retain a length of nine. Each pathway consequently yields

$$H_k \in \mathbb{R}^{9 \times 32}.$$

The three sets of local patterns are joined while preserving their sequence positions. We deliberately avoid reducing the peptide to a position-free maximum or average, because the same short motif can have a different meaning at a different position:

$$v = \text{Flatten}[H_2; H_3; H_5] \in \mathbb{R}^{9 \times 32 \times 3}.$$

The resulting vector is passed through a two-layer multilayer perceptron,

$$h = \text{MLP}(v) \in \mathbb{R}^{128},$$

Each hidden layer has 128 units with rectified linear unit activation. In the selected Human configuration, dropout with probability 0.35 is applied after the first hidden layer. Flattening the position-specific feature maps allows identical local patterns at different peptide positions to contribute differently to the representation. The multi-kernel encoder is tailored to peptides of this single length; convolution itself is not presented as a methodological novelty.

5.3. Information Sharing Across All Combinations and Within HLA Alleles

The global pathway allows every tissue–HLA combination to contribute to a shared peptide representation. During training, auxiliary tissue and HLA classification objectives encourage this representation to retain biologically relevant grouping information. These objectives do not provide gene expression, protein abundance, or other molecular measurements. If $c_i^{(t)}$ and $c_i^{(m)}$ denote the tissue and HLA categories, respectively, the global objective combines BCE with categorical cross-entropy, denoted CE:

$$\mathcal{L}_{\text{global}} = \mathcal{L}_{\text{BCE}} + \lambda_t \mathcal{L}_{\text{CE}}(\hat{c}^{(t)}, c^{(t)}) + \lambda_m \mathcal{L}_{\text{CE}}(\hat{c}^{(m)}, c^{(m)}),$$

with training-CV-selected weights

$$\lambda_t = \lambda_m = 0.3.$$

The tissue and HLA predictions guide representation learning but do not enter the final peptide score. They provide auxiliary supervision rather than independent biological evidence.

The HLA-specific pathway shares information only among tissues with the same allele, preserving allele-associated peptide preferences that broad pooling could blur. For each HLA allele m , we define

$$\mathcal{T}_m = \{(t, m) \in \mathcal{T}\}.$$

A separate encoder is trained on rows belonging to $\mathcal{T}_m$, and each tissue–HLA combination in the group receives its own linear output layer. The objective for allele m is

$$\mathcal{L}_m = \frac{1}{|\mathcal{D}_m|} \sum_{i \in \mathcal{D}_m} \mathcal{L}_{\text{BCE}} \left(g_{\tau_i}^{(m)}(h_i^{(m)}), y_i \right).$$

This pathway has no auxiliary tissue or HLA classification objective. Because every example handled by one allele-restricted model shares the same allele, the pathway concentrates on allele-specific peptide patterns and variation among tissues under that restriction. It complements the global pathway: broad sharing can stabilize combinations with few examples, whereas allele-restricted sharing can preserve motif structure that unrestricted pooling may blur.

5.4. Combining the Two Within-combination Orderings

The two pathways are trained separately, so equal raw scores need not have the same calibration in both. We therefore combine their rankings rather than their raw scores. Within each tissue–HLA combination, every candidate is converted to a percentile rank. For combination τ , let $q_{g,i}$ and $q_{h,i}$ be the two scores for row i :

$$\begin{aligned} r_{g,i} &= \text{PercentileRank}_{\tau}(q_{g,i}), \\ r_{h,i} &= \text{PercentileRank}_{\tau}(q_{h,i}). \end{aligned}$$

The equal-weight rank-fused prediction for one training run is

$$s_i^{(b)} = \frac{r_{g,i} + r_{h,i}}{2}.$$

Both pathways receive equal weight; no tissue or test set is used to choose a preferred pathway. This procedure answers a ranking question, not an absolute probability question. A peptide’s percentile depends on the other candidates scored in the same batch, so adding or removing candidates can change its final rank. Prospective use would therefore require either a predefined candidate set or a prespecified reference distribution.

5.5. Averaging Independent Training Runs and Its Scope

Neural-network training can vary across random initializations. We therefore repeat the complete training procedure three times and average the resulting rankings [15]:

$$s_{i,\text{final}} = \frac{1}{B} \sum_{b=1}^B s_i^{(b)}, \quad B = 3.$$

For each training run, the probabilities from both pathways are converted to within-combination ranks and combined.

The occurrence-matched evaluation uses the fixed internal test set. Seed-level metrics are summarized by their arithmetic mean and sample standard deviation; a cross-seed prediction average is used only where a result is explicitly labeled as an ensemble.

The averaged prediction is a single point estimate and should not be confused with training stability. We therefore report results from the individual runs as well as their prediction average, separating variation across initializations from the benefit of averaging.

5.6. Species-specific Mouse Models

The mouse analysis tests whether the same prediction problem is learnable in a second species; no human parameters are transferred to mouse. Mouse implementations are trained independently on the 11 retained tissue–H2 combinations and include simple sequence baselines, a shared encoder, H2 identifier and 34-residue pseudo-sequence conditioning, an identifier–pseudo-sequence hybrid, and matched dual-branch models with and without auxiliary tissue and H2 supervision. The selected Mouse TissuePMHC configuration uses convolution widths 3 and 5, dropout 0.35, tissue and H2 auxiliary-loss weights 0.05 and 0.10, equal rank-fusion weights, a learning rate of 3×10^{-3} , and task-balanced sampling with 32 rows drawn per task in each balanced batch. It is trained for 25 epochs. The four H2 pseudo-sequences are taken from the versioned NetMHCpan 4.1b MHC resource [6] and mapped from its H-2-* aliases to the benchmark’s H2-* identifiers; the source-file hash is retained with the derived table. All mouse methods use the same fixed-split protocol and the same three seeds as the human experiments.

5.7. Evaluation Questions

The occurrence-matched experiments ask whether peptide sequence retains tissue-associated signal after matching the MHC, source protein, and total positive tissue-occurrence count; how alternative model structures compare on the fixed Human benchmark; whether the same prediction problem is learnable in Mouse; and how much signal is recoverable without tissue identity. The Human–Mouse analysis concerns replication of the benchmark definition, not transfer of one model between species.

5.8. Fixed-Test Design and Biological Scope

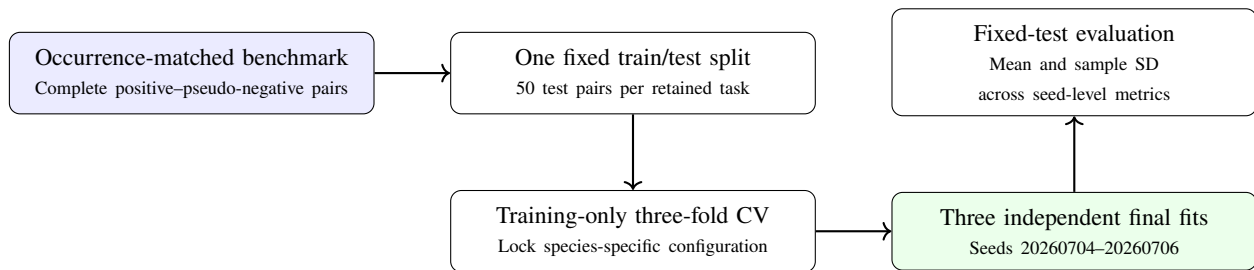
The occurrence-matched experiments use one final evaluation protocol: a fixed internal train/test split with complete matched pairs kept intact. Hyperparameters are selected by three-fold, task-stratified, pair-grouped cross-validation confined to the training file; these folds are used only for model selection and are not reported as a generalization result. After each species-specific configuration is locked, three independent models are fitted on the complete training file and evaluated on the fixed test file. Table 5 separates model selection, final fitting, and fixed-test evaluation.

Figure 4 summarizes the single evaluation pathway used for all reported results.

What is actually predicted. The benchmark is constructed from matched peptide pairs, but the main neural models do not receive two peptides simultaneously. They

Table 5: Evaluation protocol used for the occurrence-matched experiments. The fixed data split and repeated model fitting are distinct operations.

Reader-facing setting	Layer	What is done	Purpose and relationship to the other setting
Fixed internal test set	Data setting	For each retained tissue–MHC combination, 50 complete positive–pseudo-negative pairs are kept outside the training file. Every trained model is evaluated on this same fixed set.	Internal benchmark for familiar tissue–MHC contexts. It is not cross-validation and is not an external cohort.
Training-only hyperparameter selection	Model-selection setting	Three task-stratified, pair-grouped folds are formed only within the training file. Human configurations are ranked by task-macro out-of-fold AUROC with Worst-10 AUROC as the tie-breaker; Mouse configurations are ranked by the mean of task-macro AUROC and AUPRC.	Locks one species-specific TissuePMHC configuration before the fixed test is opened. These out-of-fold scores are not used as evidence of strict peptide, tissue, allele, study, or cohort generalization.
Three independent training runs	Model-estimation procedure	Each trainable method is fitted three times on the same training file with seeds 20260704, 20260705, and 20260706, then evaluated on the same fixed test file. Reported stability summaries use the arithmetic mean and sample standard deviation of seed-level metrics.	Quantifies sensitivity to initialization. The three runs are technical repetitions, not independent biological samples and not three test splits.

**Figure 4:** Evaluation workflow for the occurrence-matched experiments. Human and Mouse use the same fixed-split design and the same three final-training seeds. Cross-validation is confined to the training file and serves only to lock hyperparameters; all reported performance results use the untouched fixed test.

score one peptide row at a time using its sequence and tissue–MHC identity. Pair identifiers are used to create balanced recorded-positive and pseudo-negative examples and to keep the two rows of a pair in the same data partition. Training applies a separate present/absent loss to each row, and AUROC and AUPRC compare all positive and pseudo-negative rows within a tissue–MHC combination. Within-pair ordering accuracy asks the complementary question of whether the recorded-positive peptide receives the higher score. Keeping both members of a pair together protects data integrity; it does not turn the main training objective into a direct two-peptide ranking loss.

Saved internal test set. For each retained tissue–MHC combination, the supplied split assigns exactly 50 complete pairs to the test file and all remaining pairs to the training file. Because a peptide is used at most once within a combination during pair construction, a test peptide is absent from training rows for that same combination. The sequence may nevertheless occur in another fitting combination and can then influence shared model parameters. This test set therefore measures internal discrimination in already represented bi-

ological contexts while allowing information sharing across combinations. Human and mouse use this same rule.

Independent training runs and reported uncertainty.

Every trainable human and mouse method is fitted three times with seeds 20260704, 20260705, and 20260706. All three fits use the same species-specific training file and are evaluated on the same fixed test file. Unless a table explicitly identifies a prediction ensemble, the reported central value is the arithmetic mean of the three seed-level metrics and the accompanying variation is their sample standard deviation. The seeds quantify optimization sensitivity only; they are not biological replicates and do not increase the number of evaluated tissue–MHC combinations.

Scope of the biological conclusion. The fixed internal evaluation asks whether peptide sequence helps distinguish recorded positives from occurrence-matched pseudo-negatives in familiar tissue–MHC contexts. It does not evaluate a new tissue, MHC allele, tissue–MHC combination, study, or external cohort. Exact peptides and parent proteins may recur across different combinations on opposite sides of

the fixed split. Training-only cross-validation selects hyperparameters but does not supply evidence for ordinary same-pool or peptide-separated generalization.

5.9. Training and Baseline Models

All neural encoders and combination-specific output layers are optimized with AdamW [16]. The selected Human TissuePMHC configuration is trained for 20 epochs with batch size 512, learning rate 2×10^{-3} , weight decay 10^{-4} , dropout 0.35, auxiliary tissue and HLA weights of 0.30, and gradient-norm clipping at 1.0. The selected Mouse configuration is trained for 25 epochs with task-balanced batches, learning rate 3×10^{-3} , weight decay 10^{-4} , dropout 0.35, and auxiliary tissue and H2 weights of 0.05 and 0.10. The global and allele-specific pathways are trained independently for each random initialization. Test rows are used solely for final prediction and evaluation.

The Human and Mouse final models use random initializations 20260704, 20260705, and 20260706. Human tuning compares the default configuration, a learning-rate change, and a combined learning-rate/dropout/auxiliary-weight configuration by three-seed out-of-fold summaries. Mouse tuning evaluates staged architecture, sampling, loss-weight, fusion, optimization, and regularization choices. In both species the winning configuration is locked from training folds before the fixed test is evaluated once. Final training follows a fixed epoch schedule without test-based early stopping. Preprocessing and combination mappings are reconstructed from each fitting partition so that held-out fold labels do not influence training.

The comparison models range from simple sequence rules to architectures that share information across tissue–MHC combinations. One-hot logistic regression provides a linear sequence baseline. A BLOSUM62 random forest incorporates established similarities among amino-acid substitutions. The shared-encoder model learns one peptide representation with a separate output for each tissue–MHC combination. Dual-branch models compare global sharing with sharing restricted to one HLA allele. All methods within a table use the same tissue–MHC combinations and data split.

The conditioned baselines add tissue and MHC identifiers, an MHC pseudo-sequence representation, or both to a shared peptide encoder. Hard-sharing baselines restrict encoder sharing by tissue or by MHC group. Selective-grouping and dual-branch systems combine a global pathway with a grouped pathway; fixed-average variants weight the two pathways equally, whereas the validation-delta and validation-softmax variants derive their branch weights from the fitting-stage validation summaries. The two-view MLP is the dual-branch model with flattened residue embeddings, and the multi-kernel TissuePMHC model replaces that peptide encoder with the position-aware convolutional encoder defined above.

The broader architecture survey also includes multi-gate mixture-of-experts (MMoE) [12], FAMO [18], DB-MTL [19], CAGrad, and task-balanced shared-head training. These methods retain the same benchmark rows, task heads, seeds, and fixed test while changing the sharing or multi-task optimization rule. The pair-ranking control replaces inde-

pendent row classification with a within-pair ranking objective. These model definitions precede the result tables so that every reported system is identified before its performance is presented.

The BLOSUM62 random forest [17] concatenates the 20 substitution scores at each of nine positions into 180 input features. Each tissue–MHC combination receives a separate forest of 200 trees, with square-root feature subsampling and a minimum leaf size of two. The dual-branch neural controls use 16-dimensional residue vectors, flatten the nine positions, and apply two 128-unit multilayer-perceptron layers with rectified linear activation and dropout of 0.2 after the first layer. The model with auxiliary supervision adds tissue and HLA losses weighted 0.1 to the global pathway; the model without auxiliary supervision retains the same two pathways but omits those losses. Human and Mouse use the same three random initializations, 20260704, 20260705, and 20260706. The primary stability summaries report the arithmetic mean and sample standard deviation of the three seed-level fixed-test metrics; they do not treat seeds as independent biological observations.

5.10. Performance Metrics and Statistical Analysis

Unless stated otherwise, performance estimates (including means, differences, interval endpoints, and sample standard deviations) are reported to four decimal places. Empirical percentages are reported to two decimal places, whereas conventional nominal levels such as 95% retain their standard whole-percentage form. Counts are integers with thousands separators. Exact design constants, identifiers, and hyperparameters retain the precision needed to specify them, and figure-axis tick labels may use shorter values for readability.

The main question is whether peptides reported in the target tissue rank above comparison peptides with the same MHC restriction and parent protein. AUROC is calculated separately for every tissue–MHC combination and then averaged with equal weight, preventing data-rich tissues or alleles from dominating:

$$\text{MacroAUROC} = \frac{1}{|\mathcal{T}_{\text{valid}}|} \sum_{\tau \in \mathcal{T}_{\text{valid}}} \text{AUROC}_{\tau},$$

where $\mathcal{T}_{\text{valid}}$ contains the prespecified combinations with both labels available. AUPRC is averaged in the same manner:

$$\text{MacroAUPRC} = \frac{1}{|\mathcal{T}_{\text{valid}}|} \sum_{\tau \in \mathcal{T}_{\text{valid}}} \text{AUPRC}_{\tau}.$$

Because the paired dataset has a constructed 1:1 label ratio, AUPRC is interpreted relative to the benchmark prevalence within each combination and is not compared directly with AUPRC from naturally imbalanced clinical cohorts.

An average can hide weak performance in difficult tissues, so we also report the ten lowest-performing Human tasks and five lowest-performing Mouse tasks. Within-pair ordering accuracy is the fraction of matched pairs for which the recorded-positive peptide receives the higher score. Accuracy, F1, and MCC are secondary because they require a single decision threshold and the final score is not a calibrated biological probability.

For each random initialization, metrics are calculated separately for every tissue–MHC combination on the fixed test file and then averaged with equal task weight. The three seed-level macro metrics are summarized by their arithmetic mean and sample standard deviation. A row-level prediction ensemble is reported only when explicitly named; otherwise “three-run mean” refers to a mean of metrics rather than averaged predictions.

For the paired Human full-model versus MLP figure, uncertainty is obtained by resampling complete tissue–HLA tasks 10,000 times. Repeated training seeds quantify optimization sensitivity and are not treated as independent biological samples. Because tasks may share a tissue, allele, proteins, or related peptides, these summaries are descriptive rather than patient-cohort inference.

5.11. Cross-species Interpretation and Reproducibility

Human and Mouse use the same occurrence-matched pair definition, the same 50-pair-per-task fixed-test allocation, and the same three training seeds. They differ in the number of retained tissue–MHC combinations, MHC system, and source-data composition. Human contains 77 tasks and 25,339 pairs, whereas Mouse contains 11 tasks and 2,639 pairs. The sevenfold test-size difference is therefore determined by the sevenfold difference in retained tasks, not by different test quotas. The larger difference in training size reflects the eligible pair inventory remaining after the shared biological and occurrence-matching rules. Concordant findings support replication of benchmark learnability but do not establish a species-independent molecular mechanism or a controlled species comparison.

Both species use random initializations 20260704–20260706. Cross-species comparisons emphasize above-random signal and the value of structured information sharing. Numerical differences are not interpreted as species effects because data scale, tissue and allele coverage, and model selection differ.

For reproducibility, each experiment records the fixed training and test files, training initialization, model configuration, and command-line arguments. Reports include epoch count, batch size, optimizer, learning rate, weight decay, dropout, training time, hardware, and software versions. Predictions are saved at row level with tissue–MHC combination, label, pair identifier, seed, and score.

Seed-specific settings and elapsed times are retained with the result files, and the three-seed aggregation records the seeds and the sample-standard-deviation convention. Training-only fold assignments, out-of-fold predictions, configuration rankings, and the locked contracts are also retained. These tuning folds are not presented as a strict-generalization benchmark.

5.12. Data and Code Availability

The accompanying repository contains processed benchmark tables, fixed-split row-level predictions, seed-specific configuration files, timing records, and analysis scripts. Raw material derived from the IEDB and third-party predictor assembles remains subject to original access and licensing terms; NetMHCpan executables are not redistributed.

6. Main Results

6.1. Primary Human Benchmark

On the occurrence-matched fixed Human test set, tuned TissuePMHC reaches mean AUROC/AUPRC values of 0.8136/0.8126 across 77 tissue–HLA tasks. Accuracy is 0.7456, MCC is 0.4918, and the mean AUROC of the ten most difficult tasks is 0.6608 (Table 6).

The complete task distributions and paired TissuePMHC–MLP differences are shown in Figure 5.

Table 6: Human architecture survey on the occurrence-matched fixed test, ordered by mean AUROC. All systems use the same 77 tissue–HLA tasks. Values are arithmetic means $\pm$ sample standard deviations of the three seed-level task-macro metrics for seeds 20260704–20260706, except the $\dagger$ row-level prediction aggregate. Accuracy and MCC are means over seeds; Worst-10 is computed after first averaging each task over seeds. The tuned configuration was selected by training-only out-of-fold performance before the fixed test was opened.

Model	AUROC	AUPRC	Acc.	MCC	Worst-10
Neural single-task MLP	0.6908 $\pm$ 0.0061	0.6959 $\pm$ 0.0070	0.6389	0.2795	0.5670
HLA pseudo-sequence conditioned	0.7081 $\pm$ 0.0089	0.7136 $\pm$ 0.0071	0.6570	0.3184	0.5546
Tissue-grouped hard sharing	0.7114 $\pm$ 0.0054	0.7127 $\pm$ 0.0058	0.6555	0.3138	0.5828
Conditioned tissue and HLA IDs	0.7150 $\pm$ 0.0073	0.7191 $\pm$ 0.0052	0.6623	0.3281	0.5606
HLA ID + pseudo-sequence hybrid	0.7331 $\pm$ 0.0058	0.7359 $\pm$ 0.0055	0.6746	0.3530	0.5787
FAMO shared heads	0.7379 $\pm$ 0.0097	0.7410 $\pm$ 0.0105	0.6801	0.3620	0.5897
HLA-grouped hard sharing	0.7525 $\pm$ 0.0019	0.7517 $\pm$ 0.0012	0.6909	0.3843	0.6046
Selective global/HLA grouping	0.7643 $\pm$ 0.0015	0.7632 $\pm$ 0.0037	0.6994	0.4023	0.6143
Validation-softmax ensemble	0.7722 $\pm$ 0.0018	0.7729 $\pm$ 0.0035	0.7042	0.4118	0.6227
MMoE, 6 experts $\times$ width 128	0.7727 $\pm$ 0.0010	0.7745 $\pm$ 0.0019	0.7074	0.4190	0.6176
Shared encoder with task heads	0.7758 $\pm$ 0.0087	0.7759 $\pm$ 0.0053	0.7052	0.4138	0.6337
MMoE	0.7763 $\pm$ 0.0046	0.7792 $\pm$ 0.0045	0.7063	0.4167	0.6275
Validation-delta clipped ensemble	0.7785 $\pm$ 0.0017	0.7799 $\pm$ 0.0037	0.7098	0.4227	0.6284
MMoE, 4 experts $\times$ width 256	0.7816 $\pm$ 0.0023	0.7827 $\pm$ 0.0050	0.7146	0.4335	0.6357
Fixed global/HLA dual-branch average	0.7824 $\pm$ 0.0016	0.7851 $\pm$ 0.0035	0.7128	0.4287	0.6301
Tissue/HLA auxiliary supervision	0.7831 $\pm$ 0.0025	0.7840 $\pm$ 0.0003	0.7129	0.4315	0.6421
Auxiliary-global + auxiliary-HLA dual branch	0.7832 $\pm$ 0.0028	0.7836 $\pm$ 0.0026	0.7150	0.4342	0.6407
DB-MTL shared heads	0.7835 $\pm$ 0.0073	0.7800 $\pm$ 0.0101	0.7206	0.4436	0.6345
Pair-ranking objective	0.7841 $\pm$ 0.0030	0.7825 $\pm$ 0.0034	0.7187	0.4395	0.6431
Auxiliary-global + plain-HLA dual branch	0.7911 $\pm$ 0.0038	0.7916 $\pm$ 0.0054	0.7203	0.4447	0.6480
MLP dual-branch rank average †	0.8033	0.8018	0.7326	0.4656	0.6559
TissuePMHC (untuned)	0.8082 $\pm$ 0.0033	0.8081 $\pm$ 0.0032	0.7368	0.4742	0.6604
TissuePMHC (tuned)	0.8136 $\pm$ 0.0021	0.8126 $\pm$ 0.0021	0.7456	0.4918	0.6608

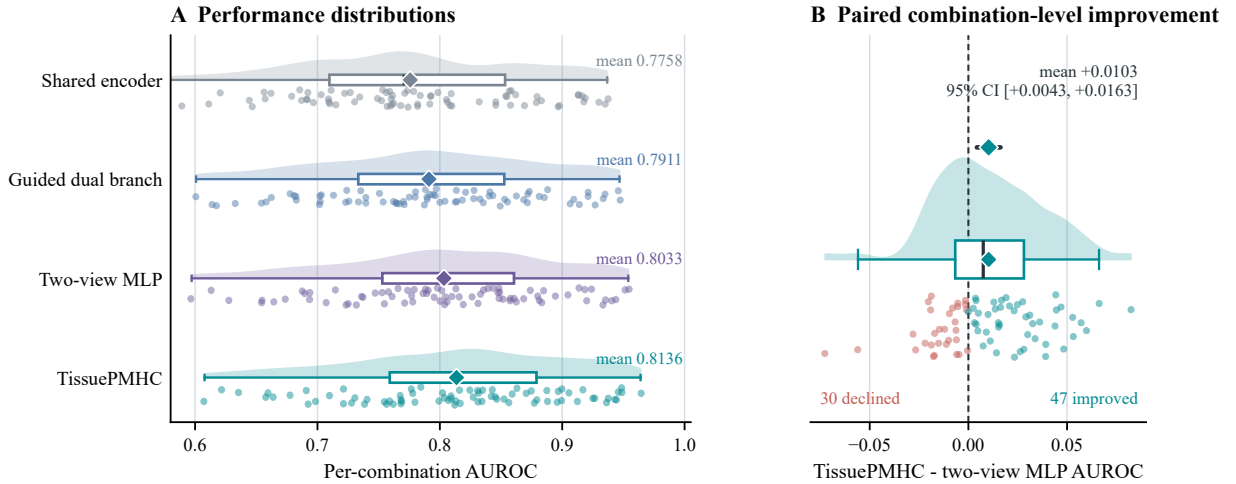


Figure 5: Human fixed-test performance on the occurrence-matched benchmark. **A**, Raincloud plots show the complete distribution across 77 tissue–HLA tasks. The shared-encoder, guided dual-branch, and tuned TissuePMHC distributions average task AUROCs over three seeds; the two-view MLP is a row-level prediction aggregate. **B**, Paired task-level AUROC differences between tuned TissuePMHC and the two-view MLP. The diamond and horizontal interval show the mean difference, +0.0103, and its 95% task-resampling interval, [0.0043, 0.0163], from 10,000 resamples; 47 tasks improve and 30 decline.

Table 7: Mouse occurrence-matched fixed-test survey across 11 tissue–H2 tasks, ordered by mean AUROC. Values are arithmetic means $\pm$ sample standard deviations of three seed-level task-macro metrics; Worst-5 is computed after first averaging each task over seeds. The tuned TissuePMHC configuration was selected by three-fold cross-validation confined to the Mouse training split before fixed-test evaluation. The row marked $\dagger$ aggregates row-level predictions and therefore has no seed-level standard deviation.

Model	AUROC	AUPRC	Worst-5
Neural single-task MLP	0.6866 $\pm$ 0.0062	0.6960 $\pm$ 0.0039	0.6012
Tissue-grouped hard sharing	0.6941 $\pm$ 0.0106	0.6866 $\pm$ 0.0142	0.6160
Conditioned tissue and H2 IDs	0.6995 $\pm$ 0.0089	0.7051 $\pm$ 0.0122	0.6209
H2 ID + pseudo-sequence hybrid	0.7102 $\pm$ 0.0065	0.7180 $\pm$ 0.0197	0.6255
H2 pseudo-sequence conditioned	0.7254 $\pm$ 0.0109	0.7361 $\pm$ 0.0033	0.6447
Selective global/H2 grouping	0.7365 $\pm$ 0.0086	0.7418 $\pm$ 0.0067	0.6308
Validation-softmax dual-branch ensemble	0.7429 $\pm$ 0.0038	0.7492 $\pm$ 0.0068	0.6372
Validation-delta clipped dual-branch ensemble	0.7470 $\pm$ 0.0039	0.7529 $\pm$ 0.0063	0.6427
Fixed global/H2 dual-branch average	0.7505 $\pm$ 0.0035	0.7552 $\pm$ 0.0046	0.6512
Shared encoder with task heads	0.7513 $\pm$ 0.0151	0.7618 $\pm$ 0.0175	0.6624
MMoE, 4 experts $\times$ width 256	0.7571 $\pm$ 0.0111	0.7668 $\pm$ 0.0190	0.6637
Auxiliary-global + auxiliary-H2 dual branch	0.7577 $\pm$ 0.0106	0.7648 $\pm$ 0.0122	0.6679
MMoE, 6 experts $\times$ width 128	0.7609 $\pm$ 0.0184	0.7672 $\pm$ 0.0189	0.6802
Auxiliary-global + plain-H2 dual branch	0.7613 $\pm$ 0.0142	0.7679 $\pm$ 0.0180	0.6637
CAGrad shared heads	0.7674 $\pm$ 0.0176	0.7796 $\pm$ 0.0179	0.6762
MLP dual-branch rank average †	0.7699	0.7793	0.6702
TissuePMHC (untuned)	0.7705 $\pm$ 0.0036	0.7785 $\pm$ 0.0031	0.6759
Task-balanced shared heads	0.7708 $\pm$ 0.0140	0.7835 $\pm$ 0.0155	0.6837
DB-MTL shared heads	0.7753 $\pm$ 0.0175	0.7880 $\pm$ 0.0179	0.6908
TissuePMHC (tuned)	0.8194 $\pm$ 0.0157	0.8279 $\pm$ 0.0121	0.7210

Relative to the three-seed MLP prediction aggregate, tuned TissuePMHC increases mean AUROC by 0.0103 and Worst-10 AUROC by 0.0049. The task-resampling interval for the mean difference excludes zero, while remaining a descriptive task-level interval from one internal fixed test rather than cohort-level inference.

survey.

Tuned TissuePMHC has the largest surveyed Mouse mean AUROC and AUPRC, reaching 0.8194/0.8279. Its seed-level variation overlaps less strongly with the previous leading DB-MTL mean than the untuned model did, but this remains a comparison on one internal fixed test rather than evidence of universal architectural superiority.

6.3. Cross-species Synthesis

After matching MHC restriction, parent protein, and total positive tissue-occurrence count, sequence-based models retain above-random discrimination in both species. Tuned TissuePMHC reaches mean AUROC/AUPRC values of 0.8136/0.8126 in Human and 0.8194/0.8279 in Mouse. In both species, the occurrence-count-only score is exactly 0.5000 in every task. The remaining signal is therefore not explained by the controlled occurrence-count shortcut, although it can still reflect antigen processing, protein abundance, cell composition, or study structure.

7. Strict Generalization and Scope

7.1. Generalization Supported by the Fixed Test

The fixed test holds out complete peptide pairs from each represented tissue–MHC task. Every test peptide is absent from the training rows of the same task, so the reported metrics measure discrimination for new task-specific rows in familiar biological contexts. The entity-overlap audit in Table 4 quantifies the cross-task reuse that remains possible.

7.2. Entity-disjoint and Unseen-context Generalization

The training-only cross-validation used for hyperparameter selection is pair-grouped within each task but does not impose global peptide or parent-protein separation across tasks. It is therefore not an entity-disjoint generalization experiment. This revision reports no numerical result for an all-

6.2. Replication on the Mouse Benchmark

The Mouse benchmark applies the same pair definition, 50-pair-per-task test quota, and three seeds without transferring Human parameters. Table 7 reports the completed fixed-test

Table 8: Completed component-related comparisons on the 77-task Human occurrence-matched fixed test. All values use the same three seeds and task inventory; the MLP row is a row-level prediction aggregate, whereas the other rows are means of seed-level metrics.

Design comparison	Systems compared	First AUROC	Second AUROC	Δ
Structured sharing	Shared encoder $\rightarrow$ auxiliary global/plain-HLA dual branch	0.7758	0.7911	+0.0153
Position-aware encoder plus tuned training configuration	MLP dual-branch aggregate $\rightarrow$ tuned TissuePMHC	0.8033	0.8136	+0.0103
Auxiliary HLA branch	Plain-HLA $\rightarrow$ auxiliary-HLA dual branch	0.7911	0.7832	-0.0079

Table 9: Human tissue-level fixed-test performance of tuned TissuePMHC on the occurrence-matched data. Each task is first averaged over seeds; tissue means then weight the available HLA tasks equally. The blank source label is reported as “NA/unspecified.”

Tissue	Tasks	AUROC	AUPRC	Tissue	Tasks	AUROC	AUPRC
Thymus	1	0.7407	0.7548	Lymphoid	23	0.8129	0.8032
NA/unspecified	1	0.7384	0.7261	Ovary	1	0.8129	0.8200
Lung	7	0.7476	0.7417	Blood	26	0.8265	0.8345
Brain	4	0.7314	0.7393	Uterine cervix	5	0.8328	0.8262
Spleen	1	0.7765	0.7834	Bone	1	0.8627	0.8762
Lymph node	5	0.8737	0.8583	Kidney	1	0.8769	0.9003
Breast	1	0.9640	0.9657				

training-combinations peptide holdout; the fixed-test overlap audit in Table 4 defines the entity reuse that remains possible.

Accordingly, the present results do not estimate performance for an unseen tissue, unseen allele, unseen tissue–MHC combination, independent study, or prospective cohort. These are separate validation targets rather than alternative interpretations of the fixed-test score.

8. Ablation and Mechanistic Analysis

8.1. Completed Component Comparisons

Table 8 reports completed fixed-test comparisons from the occurrence-matched rerun. They are not a factorial ablation: each row compares complete trained systems and therefore provides descriptive, not causal, evidence about a design change.

8.2. Complementarity of Global and Allele-restricted Sharing

On the same 77 tasks, HLA-grouped hard sharing reaches AUROC 0.7525, selective global/HLA grouping reaches 0.7643, and the fixed global/HLA dual-branch average reaches 0.7824. The auxiliary-global plus plain-HLA dual branch reaches 0.7911. These comparisons show that the occurrence-matched rerun retains useful information from combining global and allele-restricted sharing, but they do not isolate a causal contribution of either pathway.

9. Tissue and Allele Heterogeneity

9.1. Variation among Human Tissues

Performance varies across the 77 Human fixed-test tasks. For tuned TissuePMHC, the ten lowest task means have AUROC 0.6608, compared with 0.8136 over all tasks. The lowest individual task is brain–HLA-A*11:01 (0.6077), whereas the highest is breast–HLA-A*02:01 (0.9640). Table 9 summarizes tissue means. These values are descriptive: each task contains 50 test pairs, and tissues differ in allele inventory and training coverage.

The broad range should not be interpreted as a biological tissue hierarchy. Tissue identity is confounded with the avail-

able HLA restrictions, source studies, peptide inventory, and training-set size.

9.2. Variation among Mouse Tissue–H2 Tasks

Tuned Mouse TissuePMHC has task-macro AUROC 0.8194, with task means spanning 0.5016–0.9180 (Table 10). H2-Db is the weakest descriptive restriction group (0.5885), while H2-Kk and H2-Kd reach 0.9011 and 0.9018. Each restriction occurs in a different tissue subset, so these are not controlled H2 effects.

The five lowest Mouse task AUROCs average 0.7210. With only 50 test pairs per task, individual task estimates are sensitive to a small number of pair reversals.

9.3. Human HLA-level Heterogeneity

Human performance is heterogeneous across HLA loci and four-digit allele typings. For tuned TissuePMHC, HLA-A, HLA-B, and HLA-C task-macro AUROCs are 0.7778, 0.8032, and 0.8788, respectively. Table 11 reports the five lowest and five highest four-digit typings, but the loci cover different tissues and training inventories.

Errors are enriched in several low-performing HLA-A and HLA-B tasks, but no single locus or tissue accounts for all difficult cases. The fixed internal test does not separate biological heterogeneity from study, coverage, or sampling effects.

Table 10: Mouse heterogeneity of tuned TissuePMHC fixed-test performance. **A**, Task-level values, each averaged over the same three seeds. **B**, Descriptive means grouped by H2 restriction across the available tissue–H2 tasks.**A. Tissue–H2 tasks**

Tissue	H2	AUROC	AUPRC
Colon	H2-Db	0.5016	0.5025
Liver	H2-Db	0.6754	0.6695
Skin	H2-Kb	0.7821	0.8124
Liver	H2-Kb	0.7999	0.8186
Spleen	H2-Kb	0.8461	0.8524
Spleen	H2-Kd	0.8693	0.8676
Skin	H2-Kk	0.8760	0.8920
Liver	H2-Kk	0.9113	0.9126
Spleen	H2-Kk	0.9161	0.9220
Liver	H2-Kd	0.9180	0.9137
Skin	H2-Kd	0.9180	0.9440

B. H2 restriction groups

H2 restriction	AUROC	AUPRC
H2-Db	0.5885	0.5860
H2-Kb	0.8094	0.8278
H2-Kd	0.9018	0.9085
H2-Kk	0.9011	0.9089

Table 11: Lowest and highest Human four-digit HLA typings by mean fixed-test AUROC of tuned TissuePMHC. Task counts show how many tissue–HLA tasks contribute to each allele mean.

Group	HLA typing	Tasks	AUROC	AUPRC
Lowest	HLA-A*29:02	2	0.6500	0.6404
Lowest	HLA-A*11:01	3	0.6550	0.6468
Lowest	HLA-B*44:03	3	0.7036	0.6999
Lowest	HLA-B*49:01	2	0.7296	0.7105
Lowest	HLA-B*40:02	3	0.7442	0.7178
Highest	HLA-C*16:01	2	0.9351	0.9312
Highest	HLA-B*27:09	2	0.9143	0.9176
Highest	HLA-B*27:05	2	0.9120	0.9082
Highest	HLA-C*14:02	2	0.9119	0.9177
Highest	HLA-C*05:01	3	0.9010	0.8968

10. External Controls and Supplementary Results

10.1. Design of Tissue-independent Controls

The most direct biological control removes tissue identity while retaining peptide and MHC information. A previously reported trainable control of this form was not rerun on the occurrence-matched files, so its legacy performance values are excluded. The revised fixed-test comparison retains only external predictors whose scores were recomputed for every occurrence-matched test row.

Rows are not pooled, deduplicated, or assigned a consensus label after tissue identity is removed. Each original tissue-specific row and label is retained. Consequently, the same peptide–MHC query can carry both labels across tissues: it is positive where ligand evidence was reported and may be a pseudo-negative elsewhere. Table 12 quantifies this deliberate label ambiguity. A query has one retained label observation when it occurs in one row and multiple retained label observations when it occurs in more than one row; a “conflicting query” is a multiple-observation query whose rows contain both binary label values. The conflict percentage therefore uses multiple-observation queries, rather than all unique queries, as its denominator. These are not contradictions within one tissue–MHC task. They arise across tissues because the control is intentionally denied the variable that distinguishes those contexts.

The occurrence-matched control set contains 29,662 unique Human peptide–MHC queries across 29 HLA alle-

les and 3,338 Mouse queries across four H2 restrictions. We score every test query with MHCflurry and NetMHCpan, two established predictors that do not use target-tissue identity [7, 6]. These fixed external scorers are not retrained on TissuePMHC labels and have no seed-to-seed training variation.

Among queries represented by multiple tissue-specific rows, 95.18–99.04% contain both labels across the six species–partition summaries. The high conflict fractions show why a trainable tissue-blind control is intrinsically ambiguous: identical observed inputs can correspond to different tissue-specific targets. This table is a data audit, not a trainable-control performance claim.

10.2. External Tissue-agnostic Predictors and Score Combination

The occurrence-matched fixed test makes tissue-independent external scores approximately random in both species (Table 13). To integrate the former external-only and combined-score analyses in one comparable table, we also evaluate a fixed, untrained combination rule. Within each tissue–MHC task and each TissuePMHC seed, the TissuePMHC and external scores are separately converted to percentile ranks and then averaged with equal weight. No weight, threshold, or predictor is selected on the fixed test. The combined metrics and their sample standard deviations are calculated across the same three TissuePMHC seeds; the external-only metrics are deterministic. This is a sensitivity analysis of score fusion, not a separately tuned model.

All three external scores remain near random, and equal-weight fusion reduces AUROC by 0.0948–0.1102 in Human and 0.1011–0.1042 in Mouse relative to TissuePMHC alone. Thus, the general peptide–MHC scores neither explain the tissue-conditioned ranking nor improve it under this prespecified fusion rule. The complete architecture comparisons are reported once, in Tables 6 and 7, rather than repeated in a supplementary table.

Table 12: Label-conflict audit for controls without tissue information. Queries are defined by peptide sequence and MHC restriction. “One label” denotes one retained labeled row; “multiple labels” denotes more than one retained labeled row, whether or not their binary values agree. A conflicting query is a multiple-label query containing both 0 and 1. Conflict percentages use multiple-label queries as the denominator.

Species	Partition	Rows	Unique queries	One retained label	Multiple retained labels	Conflicting queries	Conflict among multiple (%)
Human	Training pool	42,978	27,063	12,572	14,491	14,168	97.77%
Human	Fixed internal test	7,700	7,012	6,327	685	652	95.18%
Human	Complete benchmark	50,678	29,662	10,441	19,221	18,882	98.24%
Mouse	Training pool	4,178	2,934	1,735	1,199	1,175	98.00%
Mouse	Fixed internal test	1,100	996	892	104	103	99.04%
Mouse	Complete benchmark	5,278	3,338	1,487	1,851	1,833	99.03%

Table 13: Unified external-control analysis on the occurrence-matched fixed tests. All metrics are task-macro. “Tissue” is tuned TissuePMHC alone; “combined” is the unweighted mean of within-task percentile ranks from TissuePMHC and the stated external score. Combined values are means $\pm$ sample standard deviations over TissuePMHC seeds 20260704–20260706. Δ AUROC is combined minus TissuePMHC for the corresponding seed, averaged over seeds. Pair accuracy is the fraction of complete matched pairs with a higher score for the recorded positive.

Species	External control	External AUROC	External AUPRC	Tissue AUROC	Combined AUROC	Δ AUROC	Combined AUPRC	Combined pair acc.
Human	MHCflurry presentation score	0.5044	0.5299	0.8136	0.7188 $\pm$ 0.0022	-0.0948	0.7032 $\pm$ 0.0008	0.7241 $\pm$ 0.0044
Human	NetMHCpan binding-affinity rank	0.4874	0.5144	0.8136	0.7034 $\pm$ 0.0016	-0.1102	0.6818 $\pm$ 0.0013	0.7057 $\pm$ 0.0022
Human	NetMHCpan eluted-ligand rank	0.4948	0.5193	0.8136	0.7101 $\pm$ 0.0017	-0.1035	0.6901 $\pm$ 0.0017	0.7145 $\pm$ 0.0031
Mouse	MHCflurry presentation score	0.5009	0.5266	0.8194	0.7183 $\pm$ 0.0115	-0.1011	0.7188 $\pm$ 0.0067	0.7303 $\pm$ 0.0141
Mouse	NetMHCpan binding-affinity rank	0.4917	0.5249	0.8194	0.7153 $\pm$ 0.0106	-0.1042	0.7055 $\pm$ 0.0055	0.7297 $\pm$ 0.0076
Mouse	NetMHCpan eluted-ligand rank	0.4924	0.5245	0.8194	0.7169 $\pm$ 0.0104	-0.1026	0.7079 $\pm$ 0.0039	0.7339 $\pm$ 0.0115

11. Discussion

11.1. Principal Findings

This study formulates tissue-specific MHC-I presentation as a shared-learning problem over peptide pairs matched by MHC restriction and parent protein. Rather than asking whether a peptide binds to or is presented by an MHC molecule in general, the benchmark asks which member of a matched pair has stronger evidence in a particular tissue–MHC combination. This formulation controls two major sources of trivial discrimination and focuses evaluation on tissue-conditioned relative evidence.

Three findings carry the main biological message. First, peptide sequence contains tissue-associated information after controlling MHC restriction, source protein, and total positive tissue-occurrence count. Second, the same fixed-test formulation is evaluated independently in Human and Mouse, although their data scales and task inventories differ and do not constitute a controlled species comparison. Third, tissue-independent peptide–MHC controls test how much signal remains without the tissue identifier. The sequence-only models cannot distinguish tissue expression, antigen processing, cell composition, and study structure.

The current occurrence-matched experiments use one fixed internal train/test split. Pair-grouped folds within the training file are used to select hyperparameters, but no peptide-separated or other entity-disjoint evaluation is reported. The study therefore does not support an estimate of the effect of excluding exact peptide identities across all combinations. The fixed test evaluates new pairs in familiar tissue–MHC combinations while allowing a peptide to recur in another training combination; it does not evaluate a new tissue, new allele, or independent cohort.

The completed mouse rerun shows useful but heterogeneous signal. Tuned TissuePMHC has the highest surveyed mean AUROC/AUPRC (0.8194/0.8279) and varies from 0.5016 to 0.9180 across the 11 combinations; DB-MTL reaches 0.7753/0.7880. The provenance-verified H2 pseudo-sequence model reaches 0.7254/0.7361 and its H2 identifier–sequence hybrid reaches 0.7102/0.7180; neither explains the advantage of the stronger multi-task structures. Because the winning TissuePMHC configuration was selected independently within Mouse training data and all final models share one fixed test, the survey supports learnability and a strong tuned result without establishing universal architectural superiority.

11.2. Relationship to General Peptide–MHC Prediction

TissuePMHC is not a replacement for general peptide–MHC prediction. General tools estimate whether a peptide is compatible with an MHC molecule or likely to be presented overall. TissuePMHC estimates tissue-associated preference relative to a matched comparison peptide. On the occurrence-matched fixed tests, the primary MHCflurry and NetMHCpan modes are near random in Human (AUROC 0.4874–0.5044) and Mouse (0.4917–0.5009). General peptide–MHC compatibility therefore does not explain the tissue-conditioned rankings observed here.

This distinction limits causal interpretation. Tissue-associated sequence patterns may reflect source-protein expression, antigen processing, cell composition, experimental sampling, or residual study structure. Because the model receives only peptide sequence and a tissue–MHC identifier, it cannot identify which process produced an association.

11.3. Potential Applications

The benchmark may support several research applications. Tissue-conditioned scores could help prioritize candidate antigens whose presentation evidence is concentrated in a target tissue, rank potential normal-tissue off-targets during early therapeutic screening, and select peptides for follow-up immunopeptidomics experiments. The same framework could also contribute to candidate triage for T-cell receptor therapies, vaccines, and other immunotherapies when used alongside binding affinity, immunogenicity, expression, safety, and population-coverage evidence.

These are potential uses rather than validated clinical applications. The balanced benchmark does not reproduce clinical prevalence, and retrospective database associations cannot establish safety or therapeutic benefit. Deployment would require prospective validation, calibrated decision thresholds, uncertainty estimates, and representative cohorts. A low predicted score must not be treated as evidence that a peptide is absent from normal tissue.

11.4. Limitations

The benchmark has several important limitations. The negative member of a pair is a pseudo-negative based on missing database evidence, not an experimentally confirmed non-presented peptide. Incomplete tissue sampling and limited detection depth can therefore convert unobserved positives into apparent negatives. This issue is especially relevant for peptides reported across multiple tissues and for tissues with heterogeneous cellular composition.

The source data are observational and inherit biases from studies in the IEDB. The revised pairing removes the specific shortcut caused by unequal total positive tissue-occurrence counts: the occurrence distributions are identical between labels within every task, and an occurrence-count-only score has AUROC 0.5000 throughout. This control does not remove uneven tissue and allele coverage, the identities of the other reporting tissues, sample preparation, mass-spectrometry platform, reporting practice, detection depth, study, laboratory, donor, or batch structure. The deliberately balanced 1:1 benchmark supports controlled ranking comparisons but does not represent real-world prevalence; its AUPRC and score thresholds should not be transferred directly to deployment settings.

Source-tissue text is retained exactly as recorded in the IEDB. Synonyms, anatomical subregions, and differently formatted labels are not normalized to a controlled ontology, so semantically related tissues may remain separate. Resolving those labels would change the benchmark and require re-training all models.

The present work covers unmodified canonical 9-mer MHC-I ligands. It does not address variable-length ligands, post-translationally modified or spliced peptides, or MHC-II presentation. It also excludes tissue transcriptomics, proteomics, source-protein abundance, flanking sequence, cell-type composition, and antigen-processing features. Tissue information enters through the combination definition and parameter-sharing structure rather than direct biological measurements.

Generalization is restricted in several ways. In the fixed

internal evaluation, peptides and parent proteins can recur across fitting and evaluation data through different tissue–MHC combinations. The split does not enforce new peptide identities globally, new parent proteins, new studies, future data, or external cohorts. Moreover, combination-specific output layers restrict evaluation to previously represented tissue–MHC combinations; unseen tissues and alleles are not tested.

The fixed test sets are internal benchmarks and repeated retrospective model comparisons can adapt analysis choices to them in ways not captured by seed-to-seed variation. In Mouse, 66.77% of unique test peptides and 80.31% of test parent UniProt identifiers occur somewhere in training; the corresponding Human values are 68.48% and 92.42%. The split therefore neither prevents all entity overlap nor constitutes external validation.

MHCflurry and NetMHCpan provide tissue-agnostic external scores but are not guaranteed to be free of overlap with their own training data. Their pretraining may include IEDB observations or related immunopeptidomic studies, and the present fixed split cannot guarantee that test peptides were absent from external pretraining collections. The small differences among external scores are descriptive and do not establish superiority of any predictor.

Finally, benchmark scale, MHC system, data composition, and the evaluated model sets differ between species, so the cross-species results cannot identify species effects. Human has 77 retained tasks and 25,339 pairs, whereas Mouse has 11 tasks and 2,639 pairs. Both allocate 50 fixed-test pairs per task and use the same three seeds; the remaining scale difference reflects the eligible occurrence-matched inventory rather than different per-task evaluation rules.

No control based on randomly reassigned tissue labels or pathway assignments is reported. A trainable tissue-blind neural control was also not rerun on the occurrence-matched files; its legacy result is excluded. Both controls require refitting and cannot be reconstructed from archived predictions.

11.5. Future Work

Several extensions follow directly from these limitations. Data splits that exclude repeated parent proteins or studies, separate future from past observations, or use external cohorts should test whether performance survives removal of additional overlap and provenance shortcuts. Models refitted after randomly reassigning tissue labels or pathways could determine whether tissue-associated gains exceed those available from the combination structure alone. Auditable checks against external predictor training collections would strengthen interpretation of the tissue-agnostic controls.

Second, the input representation can be expanded to variable-length peptides and MHC-II ligands. Tissue transcriptomics, proteomics, source-protein abundance, flanking sequences, proteasomal cleavage features, and cell-type composition could help separate peptide-intrinsic patterns from tissue context. These additions should be evaluated incrementally under splits that prevent inappropriate overlap.

Third, positive-unlabeled learning and explicit models of observation probability may better reflect the fact that

database absence is not biological absence. Calibrated uncertainty estimates could identify tissue–MHC combinations and peptides with insufficient evidence. Representations derived from tissue and MHC features, rather than a separate learned output for every observed combination, could enable principled evaluation on unseen tissues or alleles.

Finally, controlled cross-species pretraining and adaptation could test whether learned features transfer beyond the shared benchmark definition. Such experiments require harmonized data and directly comparable baselines.

12. Conclusion

We introduce Human and Mouse benchmarks that compare peptides from the same parent protein under the same MHC restriction and predict which has stronger presentation evidence in a specified tissue. The two members of every pair are additionally matched on their total number of positive tissue occurrences; a pair-level audit confirms zero mismatches and identical occurrence distributions between labels in both species and both partitions. The experiments use one fixed internal test set with 50 pairs per retained task and three common final-training seeds. Species-specific TissuePMHC configurations are selected by pair-grouped cross-validation confined to the training files; no peptide-separated generalization result is reported. Tuned TissuePMHC reaches mean AUROC/AUPRC values of 0.8136/0.8126 on Human and 0.8194/0.8279 on Mouse. Primary tissue-independent external predictors remain near random in both species. The retained signal concerns tissue-associated peptide ranking in previously represented tissue–MHC combinations and does not identify a specific antigen-processing mechanism. Validation in unseen peptides, tissues, alleles, studies, and prospective samples remains necessary before biological or clinical application.

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